

6(a)(i)	S phase / synthesis phase / interphase ;	1
6(a)(ii)	any two from: does not have an OH group (on C3) so condensation reaction cannot occur ; phosphodiester bond cannot form between deoxyribose and phosphate of adjacent nucleotide ; is a different shape because it has H instead OH (on C3) ; shape not complementary to the active site of DNA polymerase or no, enzyme substrate complexes / ESCs, can form ;	2
6(b)(i)	any two from: (both increase K_m, therefore both) reduce the affinity (of RNA polymerase for its substrate) ; (presence of) F52 reduces the affinity for the substrate more than F47 ; ora for F47 AVP ; e.g. manipulation of data to support described trend correct link stated between K_m values and affinity	2

Question	Answer	Marks
6(b)(ii)	any three from: <i>decreased, maximum rate of reactions / rate of transcription because</i> $V_{\max}$ lower, qualified ; binding (of the aptamer) changes, the tertiary structure / shape of, active site (of RNA polymerase) ; A binding changes shape of enzyme so active site less complementary to substrate (so) substrate / (RNA) nucleotides, bind less easily to active site ; A fewer enzyme-substrate complexes form binding of aptamer prevents, substrate / (RNA) nucleotide, binding to active site ; binding prevents RNA polymerase from binding to template strand ; binding prevents RNA polymerase from forming bond between RNA nucleotides ; AVP ;	3